

Exercise 2

Liuzza et al. (2018) explained authoritarian attitudes with body odour disgust sensitivity¹.

”Authoritarianism has resurfaced as a research topic in political psychology, as it appears relevant to explain current political trends. Authoritarian attitudes have been consistently linked to feelings of disgust, an emotion that is thought to have evolved to protect the organism from contamination. We hypothesised that body odour disgust sensitivity (BODS) might be associated with authoritarianism, as chemo-signalling is a primitive system for regulating interpersonal contact and disease avoidance, which are key features also in authoritarianism.”

”The BODS [is calculated on the basis of variables for] six body odour sources (breath, axillary sweat, feet, faeces, urine and gas), each of which appears in two different contexts: internal source (own smell) and external source (the smell of an unfamiliar person). (p. 3)”

In their third study, the authors showed a positive correlation between BODS scores and support for Donald Trump, with an agenda described as resonating with authoritarian attitudes. Your lecturers decided to estimate binary choice models with the data from the article. The table below presents **logit models’** parameters. Variables used as covariates:

1. *BODS* that stood for body odour disgust sensitivity (continuous; in units; higher values mean more disgust),
2. *GENDER* decoded into dummy variables,
3. *Age* (in years),
4. dummies for education attainment (high school diploma, college without degree, associate degree, bachelor’s degree, master’s degree, professional degree, doctorate degree, and the base level: lower than high school diploma),
5. *SDO* stands for an endorsement of social dominance or inequality, such as social dominance orientation (continuous; in units). The higher the value of *SDO*, the more positive the surveyed person was about social dominance (more likely to ‘step on others to get ahead’),
6. *RWA* – the Right-Wing Authoritarianism (RWA) scale, a well-established assessment measuring authoritarian dispositions (continuous; in units). With higher values of *RWA*, a person agrees more with the right-wing authoritarianism.

The dependent variable *Trump* is binary and takes value of 1 for people who were going to vote for Donald Trump in the presidential elections, and 0 for Hillary Clinton supporters.

¹Marco Tullio Liuzza, Torun Lindholm, Caitlin B. Hawley, Marie Gustafsson Sendén, Ingrid Ekström, Mats J. Olsson, Jonas K. Olofsson, *Body odour disgust sensitivity predicts authoritarian attitudes*, R. Soc. open sci. 5 2018

	<i>Dependent variable:</i>			
	Trump			
	(1)	(2)	(3)	(4)
BODS	0.290* (0.170)	0.336** (0.167)	0.333** (0.164)	0.326** (0.164)
SDO	−0.457*** (0.104)	−0.450*** (0.102)	−0.433*** (0.099)	−0.435*** (0.099)
RWA	−0.590*** (0.114)	−0.570*** (0.112)	−0.577*** (0.112)	−0.578*** (0.111)
GenderM	−0.046 (0.263)	−0.137 (0.256)		
GenderOther	−15.033 (844.241)	−14.630 (858.126)		
Age	0.016 (0.010)	0.015 (0.010)	0.015 (0.010)	
EDU.HighSchool	−1.139 (1.352)			
EDU.CollegWODegree	−0.840 (1.327)			
EDU.AssociateDegree	−0.568 (1.344)			
EDU.BachelorDegree	−1.316 (1.322)			
EDU.MastersDegree	−1.550 (1.366)			
EDU.ProfessionalDegree	−2.351 (1.772)			
EDU.DoctorateDegree	−1.624 (1.696)			
Constant	4.252*** (1.619)	2.946*** (0.933)	2.798*** (0.885)	3.417*** (0.795)
Observations	391	391	391	391
Log Likelihood	-195.666	-199.614	-200.500	-201.683
Akaike Inf. Crit.	419.333	413.227	411.000	411.365

Note:

*p<0.1; **p<0.05; ***p<0.01

- a) Why did your lecturers decided to estimate logit model instead of simple linear regression model?
- b) Select the most appropriate model from the four presented in the table. Explain your choice.
- c) Interpret parameters of the most appropriate model. What is the effect of significant variables?
- d) Can models (1)-(4) illustrate general-to-specific procedure?
- e) Let's assume that instead of *Trump* variable we want to model *candidate*. Our new variable codes for whom a person is going to vote for in the presidential elections. It takes values in the set {Donald Trump, Hillary Clinton, Gary Johnson, Jill Stein and Darrel Castle}. What is the most appropriate kind of models in this case?
- f) Now, let's imagine that we want to build a model for a different variable named *FiscCons*. The variable codes attitudes towards fiscal conservatism. Participants of the survey were asked the question "When it comes to fiscal issues, how would you describe your political attitudes?". Participants reported their responses on a seven-point scale ranging from 1 (very conservative) to 7 (very liberal). What kind of models is the most appropriate for *FiscCons*?